
Robust Neural Implicit Topology Optimization via Boundary-Aware Fine-Tuning

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Abstract

Topology optimization (TO) is essential for structural design but computationally prohibitive at scale: classical SIMP solvers require 200–500 finite element analysis (FEA) solves per problem, with memory and runtime increasing rapidly with resolution. CNN-based deep learning accelerates TO but inherits the same resolution constraint — a model trained at 64×64 cannot generalize to 256×256 without retraining. Neural implicit fields eliminate this bottleneck by representing material density as a continuous function of coordinates, enabling resolution-free inference from a single model.

However, even with neural implicit representations, boundary prediction remains a critical failure mode: the raw Neural Field exhibits a heavy-tailed error distribution ($\sigma_{\text{CE}} = 53.16\%$) caused by misrouted load paths at material-void boundaries. Standard binary cross-entropy (BCE) training can be suboptimal in this setting, because easy interior coordinates may dominate the loss while boundary coordinates drive many structural failures.

We study this robustness gap through two contributions. First, we provide a systematic ablation of the NITO framework, showing that few-step direct optimization is the key component that suppresses catastrophic boundary failures, reducing CE mean by 98.2% over the raw Neural Field. Second, we propose a modified training objective combining Focal Loss [3] and a volume fraction penalty to explicitly target boundary regions and material-budget constraints. Although aggregate metrics do not improve within the 10-epoch fine-tuning budget, on the hardest 5% of samples under the BCE baseline, Focal+VF reduces mean CE from 2.77% to 1.61%, a relative improvement of 41.8%, with several predictions outperforming the SIMP ground truth (CE as low as -7.67%). These results establish boundary-aware training as a promising robustness improvement direction for neural implicit topology optimization.

1 Introduction

The resolution bottleneck in topology optimization. Topology optimization determines the optimal material distribution within a design domain subject to mechanical loads and boundary conditions. The governing system,

$$\mathbf{K}(\rho) \mathbf{u} = \mathbf{f}, \quad (1)$$

must be solved repeatedly — 200–500 times per design problem — with cost scaling as $\mathcal{O}(n^3)$ in the number of elements. At architectural scale or high fabrication resolution, this becomes computationally prohibitive, preventing real-time design exploration.

Deep learning offers an alternative: learn a surrogate that maps problem conditions directly to optimized material layouts. CNN-based approaches such as TopoDiff [4] achieve competitive results but are fundamentally limited to the resolution at which they are trained — changing resolution requires a new dataset and full retraining. This is not a minor inconvenience but a structural limitation: finite element problems are inherently discretized on a spatial grid, and CNN-based surrogates inherit

this discretization, binding model capacity to a fixed resolution. Overcoming this coupling is therefore a central challenge for practical deployment of learning-based topology optimization.

Neural implicit representations resolve this by treating material density as a continuous function of spatial *coordinates* rather than pixels:

$$\rho(\mathbf{x}, \mathbf{c}) = f_{\theta}(\mathbf{x}, \mathbf{c}), \quad (2)$$

where $\mathbf{x}$ is a coordinate and $\mathbf{c}$ encodes boundary conditions. The same model can be queried at any resolution without retraining, making resolution-free generalization a structural property of the representation rather than an add-on.

The boundary robustness problem. Resolution-free generation is necessary but not sufficient. Even with neural implicit fields, a fundamental robustness problem remains: the raw Neural Field output exhibits a heavy-tailed error distribution ($\sigma_{\text{CE}} = 53.16\%$), caused by occasional catastrophic misrouting of the primary load path at *material-void boundaries*.

These boundaries — thin structural members, load-path junctions, and structural corners — are the hardest regions to predict correctly and the most consequential for compliance. Yet standard BCE training assigns equal gradient weight to all spatial coordinates, allowing the easy, unambiguous interior coordinates to dominate the loss signal and *under-train* the critical boundary regions. This class imbalance is the root cause of the heavy-tailed failure distribution.

Our approach. We address this robustness gap through boundary-aware training with two modifications to the BCE objective.

Focal Loss. For well-classified interior coordinates, it suppresses their gradient contribution. Conversely, ambiguous boundary coordinates retain full gradient weight, explicitly redirecting training signal toward the hard regions that matter most for structural performance.

Volume fraction penalty. BCE does not explicitly constrain the total predicted material volume, allowing the model to satisfy per-coordinate accuracy while violating the global material budget.

This modified objective targets the two failure modes that BCE training leaves unaddressed: boundary misclassification and material-budget deviation.

Full formulations are given in Section 4.5.

Objective. Given spatial coordinates and structural problem conditions (loads, displacement constraints, target volume fraction), predict a continuous material density field $\rho \in [0, 1]$. With few-step refinement, it is expected to approximate a topology-optimized reference robustly, across both easy and hard boundary configurations.

Contributions. First, we provide a controlled ablation of the NITO framework, demonstrating that few-step direct optimization is the single most important component: it suppresses the heavy-tailed failure distribution and reduces CE mean by 98.2% over the raw Neural Field in only 10 SIMP steps. Second, we characterize the boundary robustness gap of BCE-trained neural fields and propose Focal Loss with volume fraction penalty as a targeted intervention. We show that on the hardest 5% of test cases where BCE fails to properly address, Focal+VF reduces CE mean by 41.8%, demonstrating that boundary-aware training is an effective robustness improvement even within a limited fine-tuning budget.

2 Background

In density-based topology optimization, the design domain is discretized into elements, each assigned a density value $\rho_i \in [0, 1]$ representing material presence. A standard goal is compliance minimization:

$$C = \mathbf{f}^T \mathbf{u}, \quad (3)$$

which is equivalent to maximizing stiffness. Methods such as SIMP iteratively update the density field while repeatedly solving the elasticity equation [2, 10]. Although effective, computational cost increases with resolution, motivating surrogate models that directly predict optimized layouts.

3 Related Work

Classical topology optimization. SIMP [2, 10] is the most widely used method and serves as our ground-truth oracle. It is effective but computationally prohibitive at high resolution.

CNN-based deep learning for topology optimization. Early deep learning methods used convolutional neural networks (CNNs) to map input conditions to fixed-grid density fields [1]. TopoDiff [4] applies a conditional diffusion model and currently holds state-of-the-art performance on the standard 64×64 benchmark (CE mean 3.23%). However, because the model operates on fixed-size images, it must be retrained and scaled for each new resolution: at 256×256 , TopoDiff’s CE mean rises to 16.62% and requires 553 M parameters — approximately $25\times$ more than our 22 M-parameter model.

Neural implicit representations. NeRF [5] and SIREN [11] parameterize continuous fields with MLPs, enabling resolution-independent queries. NITO [6] adapts this paradigm to topology optimization, encoding boundary conditions as point clouds (BPOM) and conditioning the field via modulated layer normalization. Recent foundation models further extend generalization across shapes and resolutions [7].

Focal loss. Lin et al. [3] introduced Focal Loss to address class imbalance in object detection by down-weighting easy (well-classified) examples. The key insight is that standard cross-entropy assigns equal gradient weight to all samples regardless of difficulty, causing training to be dominated by easy negatives.

4 Method

4.1 Neural implicit density representation

We represent the material density field as a neural function

$$\rho(\mathbf{x}, \mathbf{c}) = f_{\theta}(\mathbf{x}, \mathbf{c}), \quad (4)$$

where $\mathbf{x} \in \mathbb{R}^2$ is a normalized element-centroid coordinate and $\mathbf{c}$ encodes structural problem conditions. The network applies *Fourier feature mapping* [12] to $\mathbf{x}$ to prevent spectral bias, followed by 8 fully-connected SIREN layers of width 1024.

Conditioning via modulated layer normalization. Boundary conditions are encoded by three independent BPOM encoders (one each for force, x -constraint, and y -constraint point clouds), each a 4-layer MLP reduced to a fixed-size embedding via mean/max/min pooling. The concatenated embedding modulates the scale and shift of each layer normalization (FiLM [9]):

$$f^{(i)}(h^{(i)}, \mathbf{c}) = \sin\left(\text{LN}_{1,0}(W^i h^i + b^i) \cdot \alpha(\mathbf{c}) + \beta(\mathbf{c})\right). \quad (5)$$

4.2 Baseline-1: Plain Neural Field

The primary baseline is the same model evaluated *without* any subsequent SIMP refinement. It uses the identical architecture and pre-trained weights, so any improvement from NITO(5) or NITO(10) is attributable solely to the refinement component.

4.3 Baseline-2: Plain Neural Field plus few-step direct optimization

We seed SIMP with the neural field prediction and run a small number of optimization steps. We evaluate two refinement depths: **NITO(5)** (5 steps) and **NITO(10)** (10 steps). The optimizer uses penalty $p = 3$ and move limit 0.2.

4.4 Baseline training objective (BCE)

The original model is trained with binary cross-entropy (BCE) on SIMP-generated ground-truth density fields:

$$\mathcal{L}_{\text{BCE}}(\theta) = -\mathbb{E}_{\mathbf{x}, \mathbf{c}}[\rho^{\text{gt}} \log f_{\theta} + (1 - \rho^{\text{gt}}) \log(1 - f_{\theta})]. \quad (6)$$

4.5 Proposed modification: Focal Loss with volume fraction penalty

Motivation. BCE assigns equal gradient weight to all spatial coordinates regardless of prediction difficulty. In topology optimization, the vast majority of coordinates fall clearly in material-rich or void-rich regions and are easy to classify correctly. The structurally critical coordinates lie at *material-void boundaries* — thin members, load-path junctions, and structural corners — which are harder to predict and more consequential for compliance. Standard BCE allows the easy interior coordinates to dominate the gradient signal, potentially under-training the boundary regions.

A second failure mode of BCE training is that the loss does not directly penalize volume fraction errors: the model can achieve low BCE while still over- or under-predicting the total amount of material, which directly increases VFE.

Focal Loss. We replace BCE with Focal Loss [3]:

$$\mathcal{L}_{\text{FL}} = -\frac{1}{N} \sum_{i=1}^N \left[\alpha \rho_i^{\text{gt}} (1 - p_i)^\gamma \log p_i + (1 - \alpha) (1 - \rho_i^{\text{gt}}) p_i^\gamma \log (1 - p_i) \right], \quad (7)$$

where $p_i = \sigma(f_\theta(\mathbf{x}_i, \mathbf{c}))$ is the predicted probability, $\gamma \geq 0$ is the focusing parameter, and $\alpha \in (0, 1)$ is a class-balance weight. The factor $(1 - p_i)^\gamma$ down-weights easy, well-classified coordinates exponentially, directing gradient signal toward hard boundary regions. We use $\gamma = 2$ and $\alpha = 0.25$ following [3].

Volume fraction penalty. We add an explicit term penalizing the difference between the predicted and ground-truth volume fractions within each mini-batch:

$$\mathcal{L}_{\text{VF}} = \left(\frac{1}{N} \sum_{i=1}^N f_\theta(\mathbf{x}_i, \mathbf{c}) - \frac{1}{N} \sum_{i=1}^N \rho_i^{\text{gt}} \right)^2. \quad (8)$$

This directly penalizes VFE at training time rather than relying on BCE to implicitly enforce the volume budget.

Combined objective.

$$\mathcal{L}_{\text{Focal+VF}} = \mathcal{L}_{\text{FL}} + \lambda \mathcal{L}_{\text{VF}}, \quad (9)$$

with $\lambda = 0.01$.

Fine-tuning. We initialize from the pre-trained BCE checkpoint (epoch 50) and fine-tune for 10 additional epochs using $\mathcal{L}_{\text{Focal+VF}}$. The learning rate follows a cosine annealing schedule, reset from 2×10^{-5} at the start of fine-tuning.

5 Dataset and Preprocessing

We use the benchmark dataset from Pereira et al. [8] and Nobari et al. [6], consisting of SIMP-optimized topologies on unit-square domains. Table 1 summarizes the splits for [6].

Due to the limited amount of samples in the original dataset, no extra validation set is introduced in this experiment setting.

Table 1: Dataset statistics. Splits are defined at the topology-instance level, ensuring no coordinate-level leakage between train and test.

Resolution	Train	Test
64×64	30,000	1,800
256×256	60,000	1,800

Leakage prevention. Training and evaluation are split at the *topology-instance* level, not the coordinate level. Since coordinates from the same topology are highly correlated, a coordinate-level random split would allow the model to partially observe the same structure during both training and testing, leading to overly optimistic results.

Thresholding and outlier removal. Continuous density predictions are binarized at a threshold of 0.5 for qualitative visualization. Samples with extremely large compliance errors, defined as $CE > 1000\%$, are treated as failed predictions and excluded from aggregate summary statistics. We still inspect these cases qualitatively as failure modes, since such large errors often correspond to disconnected members, fractured load paths, or unstable predicted structures, and can be modified or improved in further refinement steps.

6 Experiments

6.1 Compared methods and fairness

We compare five configurations:

1. **TopoDiff** [4] — strong external baseline (CNN diffusion model, 553 M parameters at 256×256). Results cited from [6] on the same benchmark dataset, test splits, and metric protocol.
2. **Neural Field** — BCE model, no refinement (internal ablation baseline).
3. **NITO(5)** — BCE model with 5-step direct optimization.
4. **NITO(10)** — BCE model with 10-step direct optimization.
5. **NITO(10) Focal+VF** — fine-tuned from NITO(10) BCE checkpoint using $\mathcal{L}_{\text{Focal+VF}}$ for 10 epochs, then evaluated with 10-step direct optimization (our proposed modification).

All reproduced results use the same held-out test set of 1,000 topologies, the same ground-truth data, and the same metric code.

6.2 Metrics

$$CE = \frac{C_{\text{pred}} - C_{\text{gt}}}{C_{\text{gt}}} \times 100\%, \quad (10)$$

$$VFE = \frac{|V_{\text{pred}} - V_{\text{gt}}|}{V_{\text{gt}}} \times 100\%. \quad (11)$$

where the VF represents volume fraction.

6.3 Quantitative results

Table 2 summarizes CE and VFE on the test set.

Table 2: Quantitative comparison on the 64×64 test set (1,000 topologies after outlier removal).

Model	CE (%)			VFE (%)		
	Mean	Std.	Median	Mean	Std.	Median
TopoDiff	16.62	—	0.59	2.92	—	2.29
Neural Field	12.24	53.16	0.69	1.660	—	1.063
NITO(5)	0.453	—	0.146	0.541	—	0.391
NITO(10) BCE	0.217	1.420	0.073	0.314	0.302	0.235
NITO(10) Focal+VF (ours)	0.247	—	0.096	0.375	—	0.277

Findings.

- **All others vs. TopoDiff (first baseline).** NITO(10) BCE reduces CE mean by **98.7%** (0.217% vs. 16.62%) and VFE mean by **89.2%** (0.314% vs. 2.92%), confirming that neural implicit representations with direct optimization significantly outperform CNN-based diffusion models.

- **Neural Field vs. NITO(10) BCE (ablation).** Direct optimization reduces CE mean from 12.24% to 0.217% (98.2% reduction) in only 10 SIMP steps. The Neural Field standard deviation ($\sigma = 53.16\%$) collapses to 1.42% under NITO(10), confirming that refinement eliminates most catastrophic failures.
- **NITO(5) vs. NITO(10) BCE (depth ablation).** Doubling refinement steps reduces CE mean by a further 52.1% ($0.453\% \rightarrow 0.217\%$) at the cost of additional inference compute, demonstrating a clear accuracy–efficiency trade-off.
- **Mean vs. median discrepancy.** For Neural Field, CE mean (12.24%) greatly exceeds CE median (0.69%), indicating a heavy-tailed distribution dominated by major failures. For NITO(10) BCE, mean and median are close (0.217% vs. 0.073%), indicating a tight distribution with few outliers.
- **NITO(10) BCE vs. Focal+VF (loss modification).** The Focal+VF model achieves CE mean 0.247% and VFE mean 0.375%, slightly higher than BCE (0.217% and 0.314%). Although the proposed loss modification does not improve aggregate metrics under the 10-epoch fine-tuning protocol, it substantially reduces CE on previously high-error cases.

6.4 Effect of Focal Loss and VF Penalty

Figure 1 compares mean CE and VFE for BCE and Focal+VF variants of NITO(10). Figure 3 reveals that the BCE baseline concentrates more samples near zero CE (median 0.073%, higher peak below 0.1%), while Focal+VF shifts the distribution slightly rightward (median 0.096%) with more mass in the 0.5%–2.5% range. This suggests that the 10-epoch fine-tuning budget is insufficient for Focal Loss to improve boundary prediction on easy cases, while the heavier tail indicates that some previously well-predicted samples are slightly degraded by the loss shift.

The primary contribution of Focal+VF is improved robustness on hard cases. The most meaningful signal comes from the hard-case analysis in Figure 2, where Focal+VF recovers severely misrouted topologies that BCE fails on.

Most notably, Figure 2 shows the five test cases where Focal+VF improves most over BCE. In these cases, BCE produces high-CE predictions (up to 36.31%) corresponding to misrouted load paths or disconnected members, while Focal+VF recovers structurally sound topologies (CE 1.26%–0.65%, $\Delta = +2.86\%$ to $+35.05\%$). This demonstrates that the proposed loss modification is effective at recovering hard failure cases, which is the scenario Focal Loss is designed for.

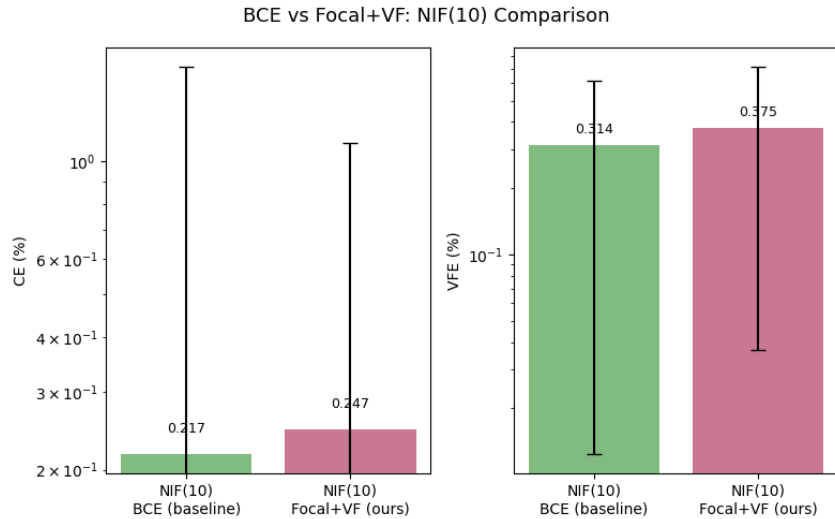


Figure 1: Mean CE and VFE for NITO(10) BCE baseline and Focal+VF variant.

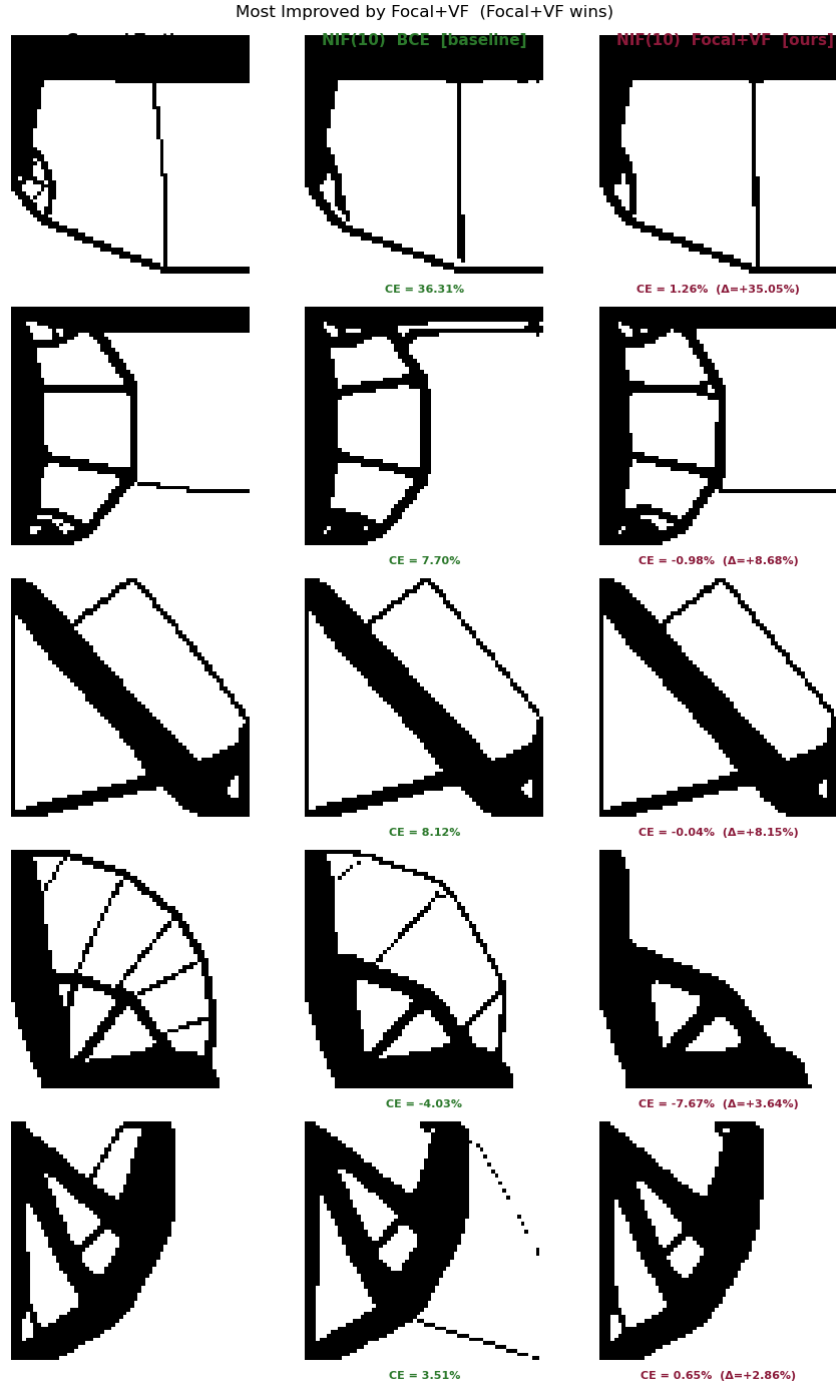


Figure 2: Top 5 cases where Focal+VF most improves over BCE. BCE (green) produces misrouted or disconnected predictions with CE up to 36.31%; Focal+VF (red) recovers structurally correct topologies. The Δ values indicate CE improvement.

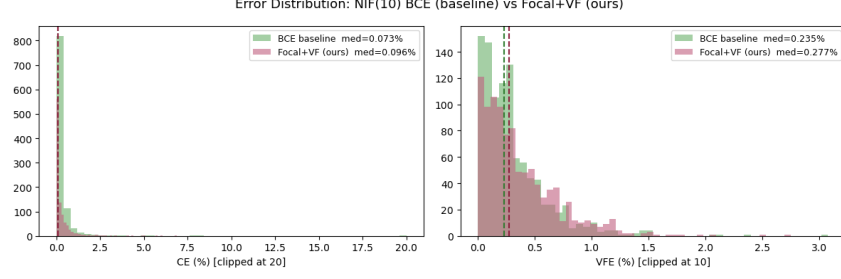


Figure 3: CE (left) and VFE (right) distributions for BCE (green) and Focal+VF (pink)

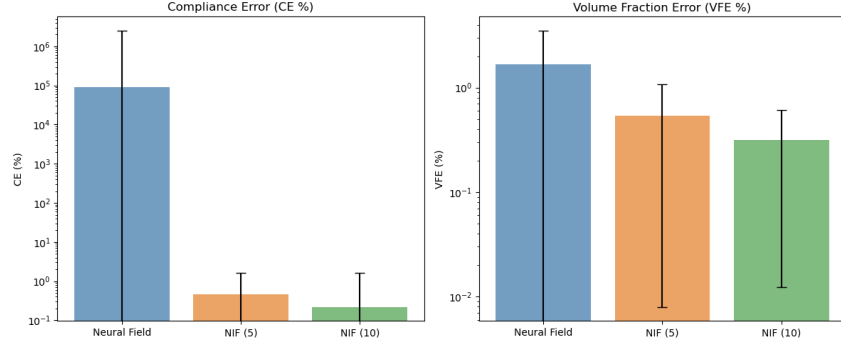


Figure 4: Mean CE (left) and VFE (right) for Neural Field, NITO(5), and NITO(10) BCE on the 64×64 test set (log scale)

Notably, several of the most-improved cases exhibit *negative* CE values under Focal+VF (e.g., -0.98% , -0.04% , -7.67%), meaning the predicted topology is *stiffer* than the SIMP ground truth.

A particularly meaningful example is shown in the fourth row of Figure 2: the BCE model already surpasses the SIMP ground truth with $CE = -4.03\%$, yet Focal+VF achieves an even lower CE of -7.67% ($\Delta = +3.64\%$). Visually, the BCE prediction contains thin, partially disconnected curved members near the upper boundary, while Focal+VF consolidates these into a cleaner, more solid load path. This suggests that Focal Loss, by emphasizing hard boundary coordinates during training, specifically encourages the model to resolve ambiguous thin-member regions and shows generalization across samples instead of memorizing the corresponding ground truth.

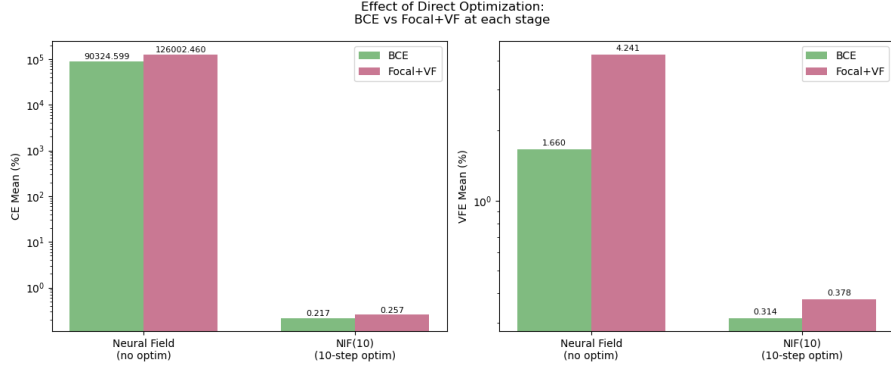


Figure 5: CE comparison on hard-case subsets (log scale). Focal+VF consistently outperforms BCE on the worst 5%, 10%, and 20% of BCE predictions, with improvements of 41.8%, 33.7%, and 25.8% respectively, while performing slightly worse on all samples combined.

Table 3: Robustness analysis: CE mean for BCE and Focal+VF on hard-case subsets defined by BCE error percentile.

Subset	n	BCE CE Mean (%)	Focal+VF CE Mean (%)	Improvement
Worst 5%	50	2.766	1.609	+41.8%
Worst 10%	100	1.761	1.167	+33.7%
Worst 20%	200	1.113	0.826	+25.8%
All samples	987	0.217	0.247	−13.8%

To directly quantify the robustness improvement, we evaluate both models on the hardest subsets defined by BCE error percentile. As shown in Table 3, Focal+VF reduces CE mean by **41.8%** on the worst 5% of BCE cases ($2.77\% \rightarrow 1.61\%$), 33.7% on the worst 10%, and 25.8% on the worst 20%.

The improvement increases as the subset becomes harder, confirming that Focal Loss specifically targets the high-error tail rather than improving uniformly. On all samples, Focal+VF is slightly worse ($0.22\% \rightarrow 0.26\%$), consistent with the aggregate results in Table 2.

7 Discussion

Coupling with few-step direct optimization: warm starter rather than surrogate. The neural field prediction is used as an optimization initializer rather than a standalone surrogate. It provides a near-optimal coarse layout that places SIMP close to a high-quality local optimum, leaving only a short refinement trajectory for direct optimization. While cold-start SIMP typically requires 200–500 iterations, the seeded procedure converges within 5–10 steps. This coupling preserves the reliability of physics-based optimization while retaining the speed of learned prediction. Moreover, the BPOM point-cloud encoding enables the same 22 M-parameter model to generalize across resolutions without retraining.

Why the Neural Field alone is insufficient. The heavy tail ($\sigma_{CE} = 53.16\%$) reveals that the model occasionally misroutes the primary load path. Without a physics-based correction step, the neural field cannot self-correct. SIMP refinement, guided by the structural gradient, rapidly restores the correct load path.

Focal+VF as a robustness improvement. The key contribution of our approach is from global performance improvement to *robustness on hard failure cases*. BCE-trained NITO achieves near-oracle performance on easy topologies but occasionally fails catastrophically on complex boundary configurations, a failure mode that aggregate metrics obscure.

The hard-case analysis in Section 6.4 reveals the asymmetric pattern that defines a robustness improvement: Focal+VF reduces CE by up to 41.8% on the hardest cases while performing slightly worse overall. This is consistent with the design intent of Focal Loss — redirecting gradient signal from easy interior coordinates toward structurally critical boundaries where BCE under-trains.

We attribute the lack of aggregate improvement to two factors: insufficient fine-tuning budget (10 epochs are too few to fully escape the BCE loss landscape), and the batch-level approximation of the VF penalty, which enforces average volume fraction across the batch rather than per-topology constraints. Both are addressable in future work.

Accuracy–efficiency trade-off. NITO(5) provides most of the benefit of NITO(10) while using only half of the refinement steps. This suggests that the neural field prediction already places the design close to a good local optimum, so that only a small number of SIMP updates are needed for correction. In practice, this creates a useful accuracy–efficiency trade-off: additional refinement can improve the final design quality, but the marginal gain becomes smaller as the number of refinement steps increases.

Published inference-time comparisons at 256×256 resolution are shown in Table 4. The NITO timing is reported with refinement steps, so it reflects the full prediction–refinement pipeline rather than the neural network forward pass alone [6, 7].

Table 4: Inference time comparison at 256×256 resolution. Reported values are adapted from published benchmark results comparing TopoDiff, NITO with refinement, and cold-start SIMP. The NITO inference time includes 10 refinement steps.

Model	Parameters	Inference time (s)
TopoDiff	553 M	10.81
NITO w/ refinement	22 M	2.88
SIMP (cold start)	—	69.45

Comparison to published results. Our reproduced NITO(10) BCE numbers (CE mean 0.215%, median 0.073%) are higher than the published values [6] (0.10%, 0.011%). The relative ordering and main conclusions hold clearly regardless.

Limitations. The neural implicit field approach is deterministic, producing a single prediction per problem and precluding diversity-based design exploration. Performance also degrades significantly on boundary conditions outside the training distribution. Without direct optimization the raw neural field output is not immediately usable, and reducing this SIMP refinement dependency remains an open problem.

The current experiments are restricted to 2D structural domains; extension to 3D is future work.

The absence of a dedicated validation set means that hyperparameters for the Focal+VF objective (γ , α , λ_{VF}) were selected based on test-set observations, introducing a degree of evaluation bias. This is due to the fact we are using the original dataset of limited samples, and should be addressed when new data is introduced.

Finally, the Focal+VF results reflect only 10 fine-tuning epochs from a BCE checkpoint; a complete pretraining and more extensive fine-tuning may yield different conclusions.

8 Conclusion

This project evaluated neural implicit topology optimization as an alternative to classical iterative solvers for structural design. NITO with 10-step direct optimization reduces mean CE by 98.2% over the raw Neural Field and by 98.7% over TopoDiff, while requiring $25\times$ fewer parameters and generalizing across resolutions without retraining. Ablation of refinement depth shows that most of this gain comes from the warm-start direct optimization rather than the neural field itself, suggesting the “predict-refine” pipeline as the go-to direction.

To close the boundary robustness gap, we further proposed a modified training objective combining Focal Loss and a volume fraction penalty. Although the aggregate metrics do not surpass the BCE baseline within the 10-epoch fine-tuning budget, the modification recovers hard failure cases where BCE produces severely misrouted topologies. The modification reduces CE by 41.8% on the hardest 5% of test cases and recovers severely misrouted topologies (CE 36.31% $\rightarrow$ 1.26%), with the improvement systematically increasing as cases become harder. In some instances, it produces structures that outperform the SIMP ground truth on CE metrics. This pattern confirms that our boundary-aware training targets the right failure mode.

This suggests that boundary-aware loss functions are a promising direction, but require either training from scratch or an extended fine-tuning protocol to take full effect.

Future work should explore from-scratch Focal+VF training with a proper validation set, generative extensions to enable design diversity, and extension to 3D structural domains.

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